Comparing Text Generating Models

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Abstract

This project explores whether algorithms can be creative when learning, based on knowledge from music lyrics. The goal is to compare the performances of two language models: one that uses a Markov Chain and a second that uses a Recurrent Neural Network (RNN). For each model, song lyrics will be generated given a user's input of genre, year, and/or artist. Performance will be evaluated using two scores. The first will be a comparison of similarities between the training data and the generated songs using a text comparison tool. The second will be a variation of the Turing test, in which human subjects will be shown lyrics and asked whether or not they think the lyrics were created by a human or a machine.

1 Introduction

1.1 Motivation

The three of us were initially motivated to pursue a project related to natural language processing and machine learning since we all speak another language other than english. We ended up deciding to explore computer-generated language, with the goal of better understanding how machines generate text. As a part of the exploration, we also were fueled with the curiosity of how to recognize machine-generated text in real life. In terms of computer-generated text, we ultimately decided to zoom in on computer-generated song lyrics, prompting bigger questions like: What makes a song a song? What metrics can be used to evaluate the computer-generated lyrics? What does it mean for something to be creative? Are computers demonstrating creativity through the computer-generated lyrics?

1.2 Related Work

The area of Natural Language Processing is a growing field that has garnered an increasing amount of attention in the past couple of years. One part of Natural Language Processing research explores computer generated language in different contexts. In the case of this study, we focus on computer generated lyrics, and explored other similar research.

One example of similar research is discussed in the paper "Stylistic Transfer in Natural Language Generation Systems Using Recurrent Neural Networks" written by Jad Kabbara and Jackie Chi Kit Cheung of McGill University. In their study, Kabbara and Cheung explore different neural network approaches "to automatically learn and incorporate stylistic features in the process of language generation" [Kabbara and Cheung, 2016]. The discussions presented in the paper, especially around stylistic features of language, are applicable to our project on song lyrics. For example, the paper discusses "three main criteria for the evaluation of stylistic transfer systems: soundness (i.e., the generated texts being textually entailed with the original version), coherence (e.g., free of grammatical error, proper word usage, etc.), and effectiveness (i.e., the generated texts actually match the desired style)" [Kabbara and Cheung, 2016]. The criteria for evaluation Kabbara and Cheung use would help in the evaluation of our outputs, particularly when considering different genres of music. It is possible that the lyrics of different genres of songs possess particular stylistic features, shown through soundness, coherence, and effectiveness, that might become apparent when generating new song lyrics.

Another example of related research is presented in "Dope Learning: A Computational Approach to Rap Lyrics Generation." The paper, written by researchers from Aalto University in Finland, delves into the intricacies of generating rap songs. Different from any old song, rap songs focus on having an interesting story and complex rhyme patterns. The unique identity of rap songs drove their research, and helped ask interesting questions about the intersection between computational creativity and information retrieval [Malmi et al., 2016]. In their research, Malmi et al. used models based off machine learning techniques: the RankSVM algorithm and a deep neural

network model with a novel structure. The output of their data is now in a website that allows users to generate their own rap lyrics based on a chosen genre. The research described in this paper is a starting point for an exploration of how computers can mirror stylistic details, like rhyming patterns. Our computer generated lyrics do not contain any stylistic features, but the discussion prompted by Malmi et al. encourages further exploration of songs and styles of certain genres.

Inspired by Malmi et al's exploration of the intersection between computational creativity and information retrieval, we also read about more in depth research about 'creative language retrieval' in a paper called "Creative Language Retrieval: A Robust Hybrid of Information Retrieval and Linguistic Creativity" written by Tony Veale from the University College Dublin. [Veale, 2011]. The paper discusses a hybrid of figurative language processing and information retrieval, discussing the definition of creativity. The complexity of this paper and the related discussions confirm the blurred lines that exist in the discussion about creativity of computer generated lyrics.

2 Methodology

2.1 Data

Our model was trained on Kaggle's database of "380,000+ lyrics from MetroLyrics." Each data entry included information about a song's title, artist, genre, year created, and lyrics. The song lyrics inputted to the models were grouped by artist (Dolly Parton) or randomly sampled to sizes of 1000, 2500, and 5000.

2.2 Algorithms

Three different models were trained to generate the lyrics: a character level recurrent neural network[1], a word level recurrent neural network[2], and a Markov chain model[3]. We chose to look deeper into RNNs because they have been used to achieve great success in various natural language processing tasks. In particular, RNNs are powerful for language generation because they are able to predict the likelihood of the next character/word given the previous characters/words, meaning the output of the trained model would be the sequence of predicted characters/words.

2.3 Performance Evaluation

To compare the computer generated lyrics, a text comparison tool[4] was used to measure the number and length of n grams the computer generated lyrics had in common with the human ones. Additionally, human volunteers participated in a survey of 6-7 questions that asked them to distinguish between human and computer generated lyrics. In both comparisons, only the performances of the char level RNN and the Markov model were compared. One focused on performance based on the Dolly Parton lyric data and the other focused on performance based on 5000 sampled songs.

3 Results[5]

In total, 70 different sets of song lyrics were generated using different models. In some of the models, the song lyrics is split up by line, but in others, the song lyrics was outputted as one long line. The chart below indicates the different lyrics produced. The bolded sections are the results that we focus on in our analysis.

	Character level RNN	*Word level RNN*	*Markov Chain*
Dolly Parton songs	**10 songs**	---	**10 songs**
1000 songs	10 songs	---	---
2500 songs	10 songs	---	---
5000 songs	**10 songs**	10 songs	**10 songs**

Table 1

3.1 Text Comparison

The figures below illustrate the average number of times a group of n consecutive words were "copied" from training data and "pasted" into the machine-generated song. In the figures, n is represented on the x axis and and the frequency of n on the y axis.

In Figure 1, the output from the Markov machine is represented. The longest set of consecutive words was 7 in this case. This is likely due to the fact that a Markov machine is dictated by the probabilistic outcome of one word following another word. It is therefore unlikely to get a long string of words "copied and pasted" from the training data. On the other hand, it is highly likely that short sequences of words will reoccur, as illustrated by the fact that there were on average 24 instances of 4-word sequences "copied and pasted."

In contrast, Figure 2 was generated by the RNN model. Here we can see the longest set of consecutive words was 12, and the frequency of 4-word sequences was only

1 https://github.com/spiglerg/RNN_Text_Generation_Tensorflow. Accessed April 3, 2018.

2 https://github.com/hunkim/word-rnn-tensorflow. Accessed May 10, 2018.

3 https://github.com/codebox/markov-text. Accessed April 3, 2018.

4 https://github.com/alimony/text-comparison-tools. Accessed April 10, 2018.

5 Our Github repository can be found at https://github.com/catocity/cs232.

4.25 (on average). This underscores the difference between the two training models.

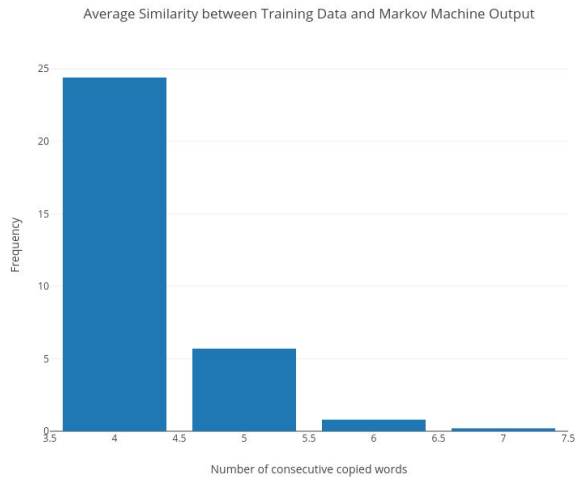

Figure 1

Figure 2

We then aggregated 5000 songs and implemented the Markov Chain model on the data to generate a new output. The results of one of the outputs is displayed in Figure 3. In contrast to Figure 1, one will notice that the average frequency of repetition is much smaller. This is likely due to the fact that the output was a shorter song.

Next, we implemented the character level RNN model on the same set of 5000 songs. The results are displayed in Figure 4. Unlike Figure 2, it is much less likely that one will find long sequences of word-for-word "copies" from the training data to the output since the model is run at the character level. Nevertheless, they do exist (up to five words in a row).

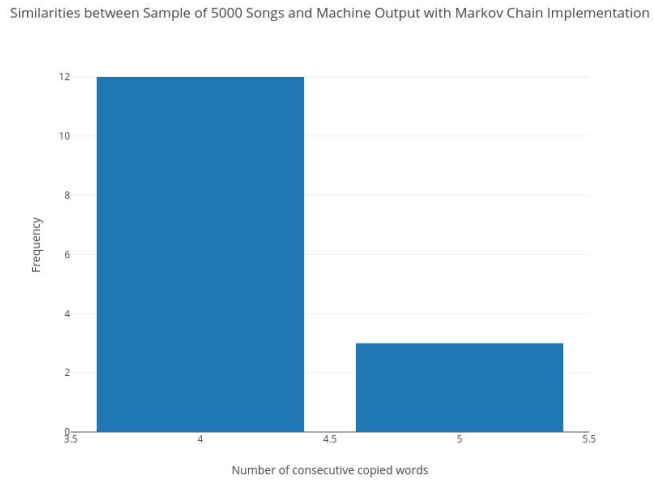

Figure 3

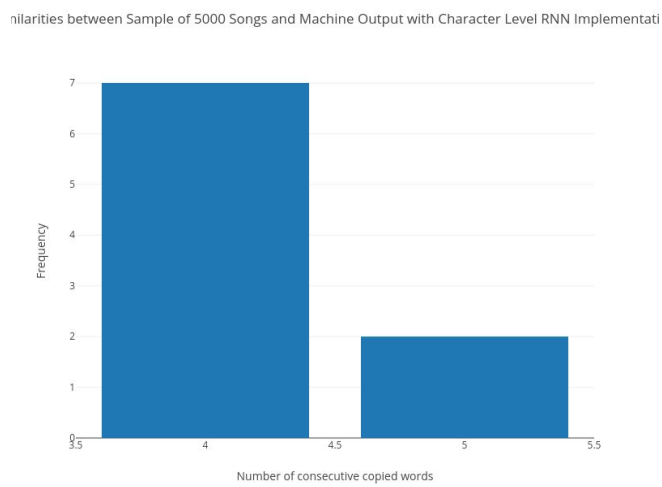

Figure 4

3.2 Human Evaluation

For our human evaluation, we released two different google forms asking users to guess whether the indicated lyric was generated by a human or a computer. One of the google forms featured lyrics generated from the Dolly Parton songs while the other google form featured lyrics generated from the 5000 songs sample set.

Both of the google forms were randomly sent out to Wellesley students, meaning the demographic of the respondents were college-aged biological women.

Figures 5 and 6 depict the results from the first survey, with the lyrics generated from the model trained on Dolly Parton songs. Figure 5 shows the responders' correctness to the human generated lyrics and Figure 6 shows the responders' correctness to the computer generated lyrics. As you can see, only 57.8% of responders were able to correctly identify human written lyrics and 61.5% were

able to correctly identify computer generated lyrics. These percentages are close to 50%, indicating that the deduction is similar to randomly guessing who created the song lyrics.

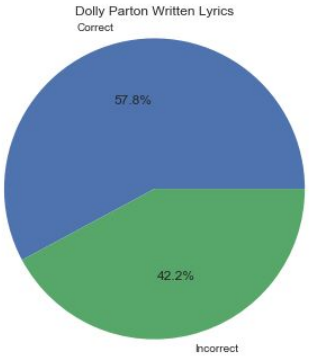

Figure 5

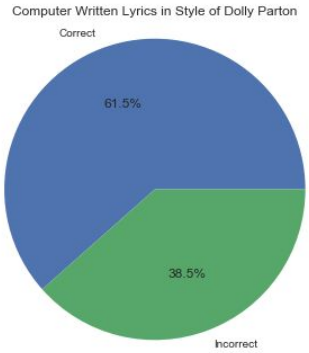

Figure 6

Figures 7 and 8 depict the results from the second survey, with the lyrics generated from the model trained on 500 sample size set of songs. Figure 7 shows the responders' correctness to the human generated lyrics and Figure 8 shows the responders' correctness to the computer generated lyrics. As you can see, only 63.3% of responders were able to correctly identify human written lyrics and 56.7% were able to correctly identify computer generated lyrics. Interestingly enough, the performance of the model trained on more data performed worse than the models trained on less. This may be attributed to the presence of different languages in the sample_5000 dataset. We would have expected the RNN to perform better when trained using more data. Another factor that may have caused this error is that the survey size was not large enough. We had 34 responders to the Dolly Parton trained model and 30 responders to the 5000 song trained model. This is not nearly large enough to accurately evaluate the performances of either models.

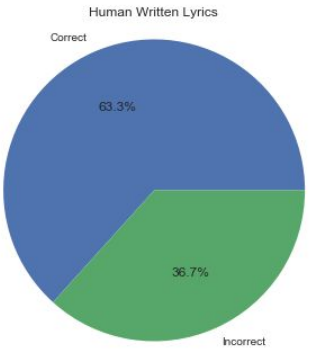

Figure 7

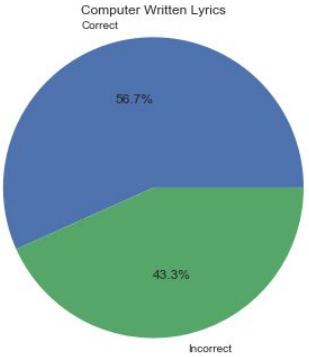

Figure 8

In addition to the above mentioned patterns about predicting human versus computer, there were also some differences between the RNN and the Markov Model for the 5000 song trained model. Figure 9 shows the votes for two different RNN generated lyrics (lyrics indicated above the graph), while Figure 10 shows the votes for two different Markov Chain generated lyrics, and Figure 11 shows the votes for two human written lyrics.

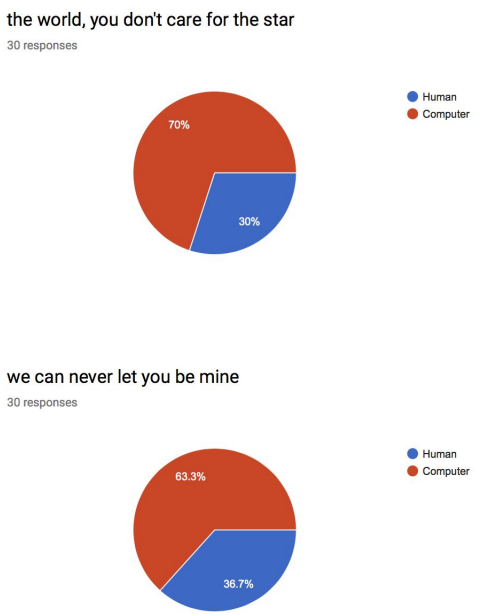

Figure 9 - RNN generated lyrics

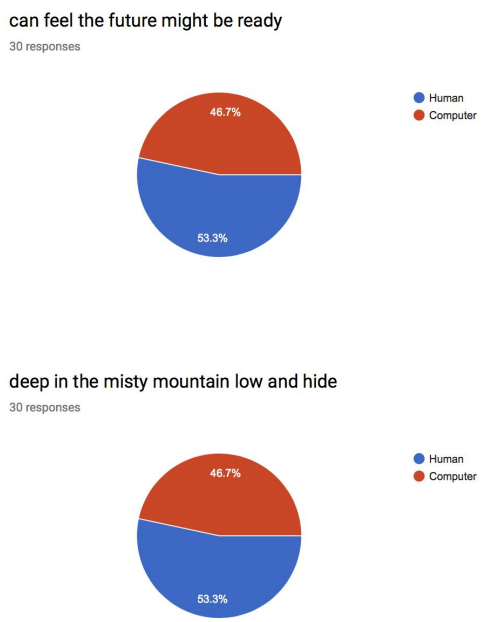

Figure 10 - Markov generated lyrics

Figure 11 - Human written lyrics

4 Conclusion

4.1 Discussion

Overall, our work is inconclusive. General research has that RNNs are able to perform better with more training because of their ability to remember past patterns and predict likelihood of future characters or words.

A particularly interesting trend on the human evaluation data is the RNN generated lyrics were decisively voted to be computer generated (70% and 63.3% for the first and second RNN test lyric, respectively). On the other hand, the Markov Chain generated lyrics recieved about 50/50 votes (46.7% and 46.7% for the first and second Markov test lyric, respectively). Lastly, for the human written lyrics, they were largely voted to be human generated (66.7% and 60% for the first and second human test lyric, respectively). The patterns in people's ability to guess if lyrics are generated by humans or computers, especially the differences between the RNN and the Markov Chain model is potentially an indication of the importance of grammar in song lyrics. As a user of the first human survey complained, one of the human generated lyrics appeared computer generated because the sentence was cut in the middle, making the lyric sample appear to have improper grammar. At what point can artistic license come in? Is there a point that rhyming and creativity could mask improper grammar?

Contrary to expected, our results do not clearly show a significant benefit to RNNs (either character based or word based), and in fact, might even show a benefit to the Markov Chain Model based on the computer evaluation.

The lack of statistically significant differences indicates that more work is needed to validate this project and correct for certain errors that were made.

4.2 Future Work

In the future, this project could be improved upon by exploring the models more in depth. Non-english languages could be filtered out of the dataset to provide a better baseline on how language generating models perform with English words. Another option would be to test the models' capabilities to perform well with foreign languages. This approach would give insight into whether biases of English are engrained in these models. Continued exploration of the impact of artist, genre, and year on generated lyrics would also be possible with the availability of more data. This paper focused more on the evaluation of Markov Chains and RNNs and the generation of human-like lyrics, but failed to answer one question that we initially posed, *what does it mean to be creative?* Further analysis of the structure of human and computer generated song lyrics may be able to offer some insight into this inquiry.

Acknowledgments

Thank you to Eni Mustafaraj for the support in the generation of this project. In addition, thank you to Scott Anderson, Lyn Turbak, and Ben Wood for helping us figure out which machine we could use to train our model overnight.

References

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